

Proposed Morphology Pipeline

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0. Imaging Parameters

- 100 μ m resolution;
- 3-second interval between frames;
- 40-frame “stacks”.

1. Quantifying “Donut-ness”

Initial approach:

1. segment image into potential donuts;
2. Find fluorescence profile of each individual candidate and discriminate “blob” vs. “donut”.

1.1. Denoised Segmentation of “Mitochondrial Objects”

1.1.1. Important Parameters

Basic filtering/denoising parameters need to be determined (similarly to Cellpose case).

- Maximum eccentricity (`maxecc`; how “round” an object needs to be);
- Minimum size (`minsize`; how large an object needs to be).

1.1.2. Watershedding

Likely will need to watershed to deal with candidate units engaging in fusion/fission events.

During initial segmentation, is worthwhile to watershed for candidates (assumed to be merged) that are size $\leq 1.5 \cdot \text{minsize}$ and $\text{ecc} > 1.7 \cdot \text{maxecc}$.¹

¹Coefficients subject to change.

1.2. Fluorescence Profile

For each object, generate cross-section ROI line across both major axes (perpendicular always). Then, find fluorescence profile for each axis. If a fluorescent “divot” is observed for both plots, the object can be considered a donut.

Need to determine the length and magnitude of the divot to make sure that drops in fluorescence are not just noise.

A rigorous definition for differentiation:

- Uniformity of “ring” width:²
 - Find the width of both sides of the fluorescing ring for both ROIs (4 total measurements);
 - For each ROI, $w_1 = w_2 \pm n_a$ for some leniency n_a .
 - Compare ROIs by looking at $\max(w_1, w_2)_{\text{axis}=0} = \max(w_1, w_2)_{\text{axis}=1} \pm n_b$ for some other leniency n_b .
- Substantiality of divot:
 - Ensure that, for each ROI, $\frac{w_1 + w_2}{(w_1 + w_2) + w_{\text{divot}}} < 0.5$.³
- Fluorescence difference of donut:
 - Want to see that $\min(\max(\text{fluor}_{l_1}), \max(\text{fluor}_{l_2})) > \max(\text{fluor}_{\text{divot}})$ for both axes.

²Extremely skewed donuts have not been observed (at least by me).

³Based on heuristic that the donut ring thickness is never more than half of the diameter of the whole object.